Jia Duan (Jade)

Applying for PhD positions in NLP, starting Fall 2027

Nanjing University of Science and Technology · M.Eng. Candidate, Cyberspace Security
Expected Apr. 2027 · jadeluckybuff@njust.edu.cn

Education

- Sep. 2024 – Apr. 2027 (expected) | Nanjing University of Science and Technology | M.Eng. in Cyberspace Security
- Sep. 2019 – Jun. 2023 | Beijing Information Science and Technology University | B.Eng. in Information Security

Research Interests

Natural language processing; reliability and interpretability of large language models; counterfactual evaluation. I study how models use task-relevant evidence, and how training and inference settings shape the reliability of their judgments and generated explanations. My application domain so far is harmful-language understanding.

Publications and Research Contributions

Three sole-first-author works: what evidence a model uses → whether evaluation metrics reflect the right behavior → whether generated explanations are reliable. For each, I led problem formulation, method and experiment design, implementation, analysis, and writing.

Accepted

[EMNLP 2026 Main] When Counterfactual Debiasing Backfires: Identity Cues as Task-Dependent Evidence in Harmful Language Understanding — First Author
Jia Duan, Xiaoyu Lu, Wenjing Hu, Zihao Ding

- Uniform counterfactual debiasing can delete evidence a task needs: an identity mention may act as a spurious shortcut for cyberbullying detection, yet it is label-defining evidence for target identification. Proposed task-aware representation routing, which improves target identification on three datasets while keeping detection and sentiment performance stable.

[ICME 2026] Behavior-driven Hypergraph Modeling for Accurate Cyberbullying Detection in Social Media — Fifth Author

- Models social behavioral relations with hypergraphs to improve cyberbullying detection.

Under review (ACL Rolling Review, October 2026 cycle)

Identity-Edit Sensitivity and False Positives in Offensive-Language Detection — First Author

- **Evaluation validity.** Does a drop in counterfactual sensitivity actually mean fewer false positives? Tested this link with controlled training interventions and found that the metric can improve without the expected change in model behavior.

Seed and Decoding Effects on Target Attribution in Implicit Hate Explanation — First Author

- **Explanation reliability.** Studied how random seeds and decoding settings change which target group LLM explainers name in implicit-hate explanations; separated changes in discrimination from changes in decision tendency, and used controlled comparisons to locate the effect of a decoding default.

Research Experience

Research Assistant, Center for Brain-Inspired Computing Research, Tsinghua University Jul. 2026 – present

- Study environment factors and dynamic structure in video world-model representations; design structured models and generalization tests.
- Compare how different data splits change the conclusions of structure diagnostics, to test whether models learn relations that generalize. Ongoing.

Technical Skills

- **Model training:** multi-task fine-tuning of BART, RoBERTa, DeBERTa; LoRA tuning of Qwen2.5-7B and Llama-3.1-8B; decoding-setting comparisons.
- **Counterfactual data and evaluation:** lexicon-based identity masking, paired counterfactual evaluation, linear probing.
- **Experiments and statistics:** multi-seed experiments, paired bootstrap confidence intervals, Wilcoxon and Cochran's Q tests.
- **Tools:** Python, PyTorch, Hugging Face Transformers, multi-GPU Linux servers.

Other

- Patent: edge–cloud collaborative method and device for multimodal emotion perception (first inventor; patent application filed).
- Contributed to an NSFC proposal on interpretable decision rationales for multimodal cyberbullying perception.
- China Graduate Electronics Design Contest 2025 (team leader): Third Prize, East China Division; First Prize, university level.
- President, university piano ensemble (undergraduate); docent, Harbin Military Engineering Institute Heritage Museum.